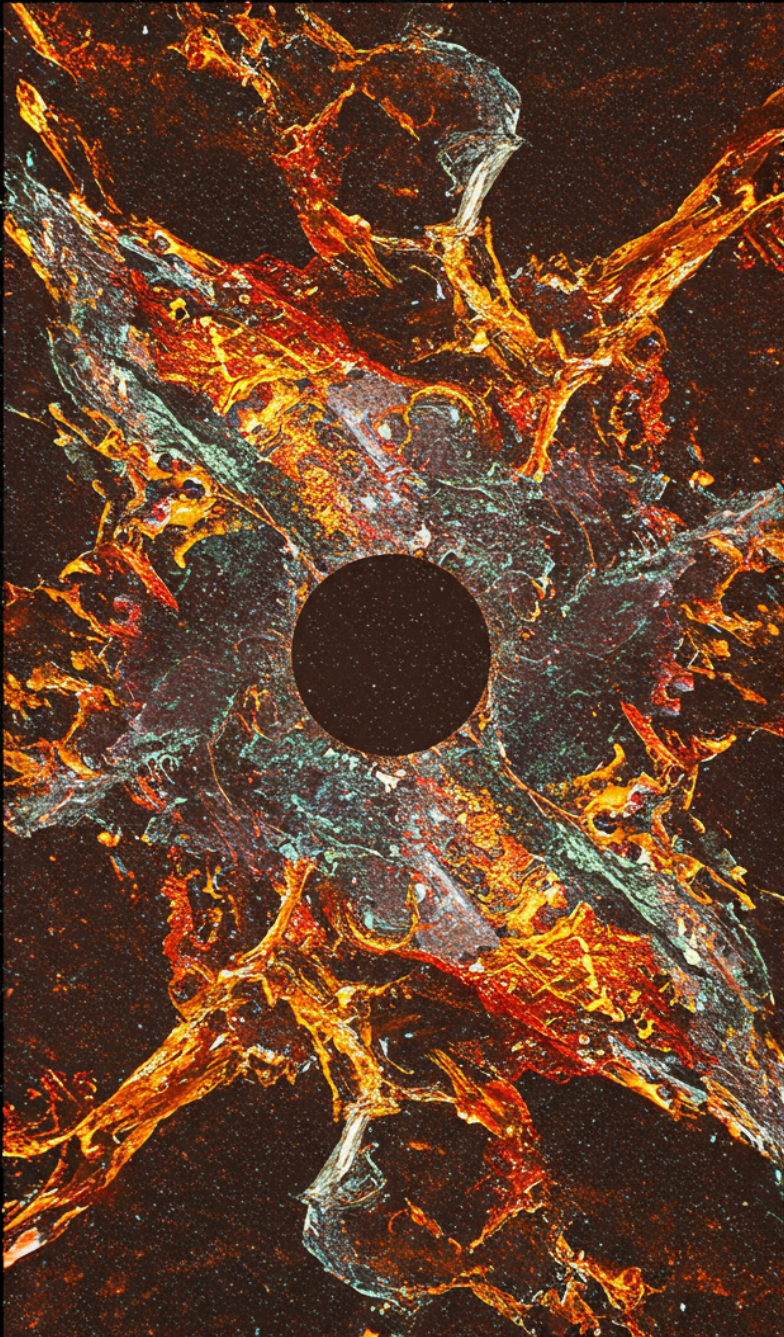




Remixing for Democratic Collective Intelligence

Design and Implementation of a Scalable Platform

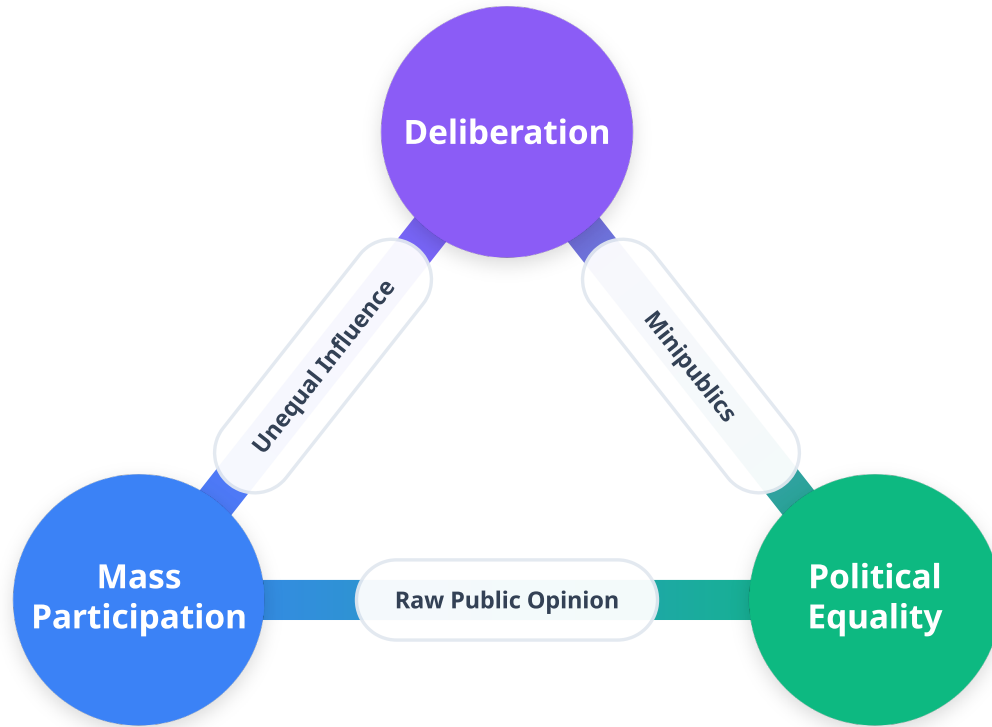
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The Democratic Trilemma

According to James Fishkin



Problem Setting

- **Deliberation:** Quality interaction, but limited to ~5–30 people
- **Delegation:** Scales, but narrows participation to representatives
- **Crowdsourcing:** Mass participation, but institution retains authority

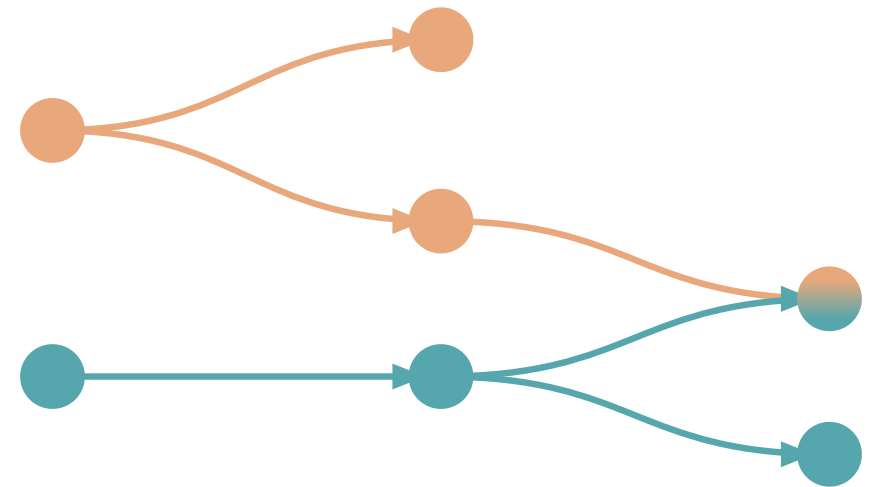
Synesis: Pushing the Limits of Democratic Participation

Remixing: Asynchronous building-upon — iterative, direct authority, scalable

From Greek *s'ynesis* ("bringing together," "understanding"): the capacity to connect scattered pieces of information into a coherent whole. The platform's core mechanism: synthesizing distributed contributions into collective outcomes.

Shared Knowledge Base: The DAG

- **Shared knowledge base** → **fairer outcomes:**
Carpentras et al. (2024) show that storing partial solutions → both majority *and* minority solutions improve
- **Ideas = Nodes, Evolutions = Directed Edges** → a Directed Acyclic Graph
- **Non-destructive:** Parent and child coexist ("Git for deliberation")



Phases



Discussion: *continuous, asynchronous*: creation and evaluation happen simultaneously, fitting people's real schedules.

Full voting Phase: The final candidates must be presented to all

The Attention Problem

With simultaneous adding and voting in an asynchronous system:

- **First-mover advantage:** Early proposals accumulate views and subscriptions before alternatives exist
- **Information overload:** N participants $\times$ P proposals = quadratic attention matrix — nobody can read everything
- **Discovery failure:** Good ideas buried under volume

The system must guide users: counteract first-mover effects, aid discovery of under-explored ideas, and sift out the best branches in the DAG so people can work on those.

Windows: Collective vs. Personal

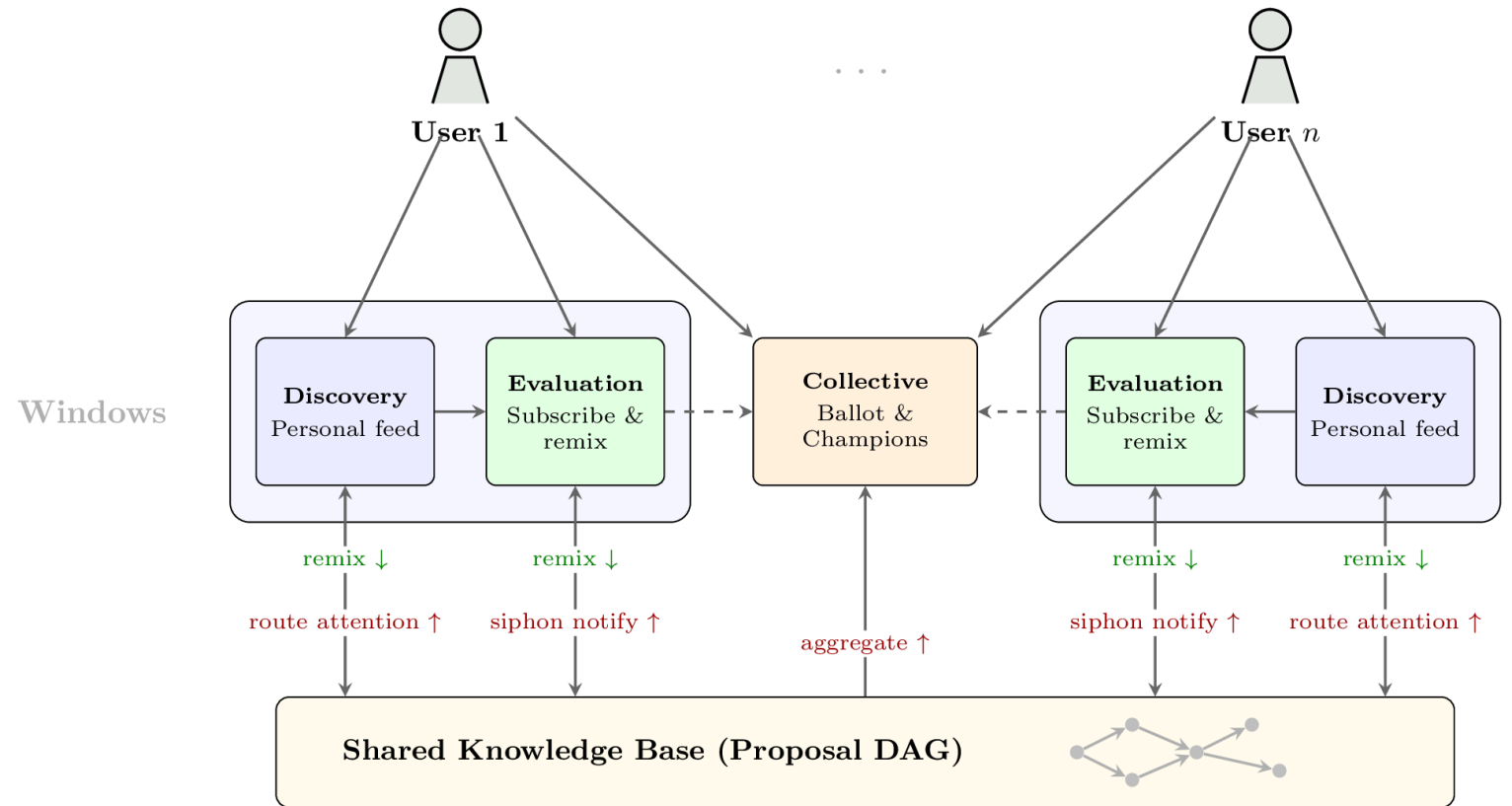
The DAG can be viewed from two perspectives, in productive tension:

Collective:

"What are the best ideas?"

Personal:

"Where can I contribute?"



Mechanisms: PersonalFocus & Siphon Effect

Instance of Discovery Window

PersonalFocus Algorithm

- **Anti-preferential attachment:** each view *reduces* future visibility → no rich-get-richer
- **Novelty premium, diversity boost, personal affinity**
- **Under-explored ideas** surface automatically

Instance of Evaluation Window

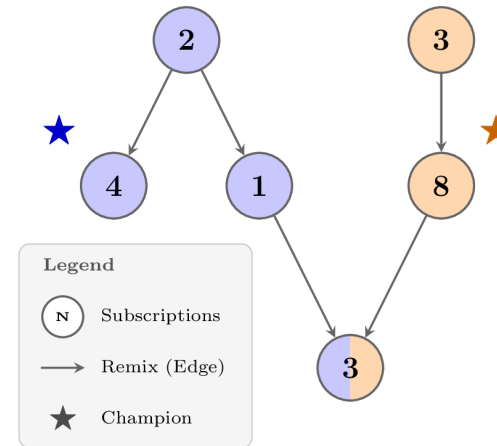
Subscriptions & Siphon Effect

- **Siphon Effect:** When a proposal is remixed, its subscribers get notified
- If they prefer the remix, they move their subscription
- **Support flows "down" the DAG** to better versions

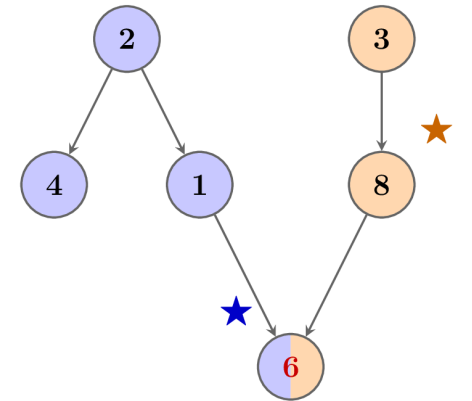
Convergence: Labels & Ballot Diversity

- **User-generated labels** categorize proposals thematically
- Labels **inherit through the DAG**: edits keep parent's label, combinations inherit the **union** of both parents' labels
- **Label-Exclusive Ballot**: up to 7 Champions, no label overlap → guarantees structural diversity

Time t_1 : Initial Merge



Time t_2 : Partial Domination



Value-Sensitive Design Choices

- **Labels** → **parallel, unintrusive discussion + diversity-preserving aggregation:**
Thematic branches evolve independently; the ballot algorithm ensures minority perspectives reach the final vote without requiring moderation
- **PersonalFocus** → **balancing group vs. individual attention:**
Anti-preferential attachment distributes collective exploration fairly, while personal affinity respects individual interests
- **AI-agnostic design:**
Every system action traceable to explicit user actions and documented formulas — no black-box ML models. Simple heuristics over opaque algorithms.
- **Privacy-first computation:**
PersonalFocus runs **entirely client-side**

Evaluation: Two Approaches



Human Experiments

- 2 community deployments (N=11, N=12)
- Community defined normative questions
- Mixed methods: telemetry, SUS survey, think-aloud



Simulation Testbed

- 90 runs, 30-day window
- Agents use **actual platform API**
- Scales to N=400
- Empirically calibrated from human data

Human Experiments: Setting

Study 1: WG Studiengruppe

- **11 participants**, 12 days
- 1 question: *fair cost-sharing for shared purchases*
- Pre-existing flat-sharing community

Study 2: WG-Netz Winterthur

- **12 participants**, 19 days
- 2 questions: *network identity + resource sharing*
- Pre-existing community network

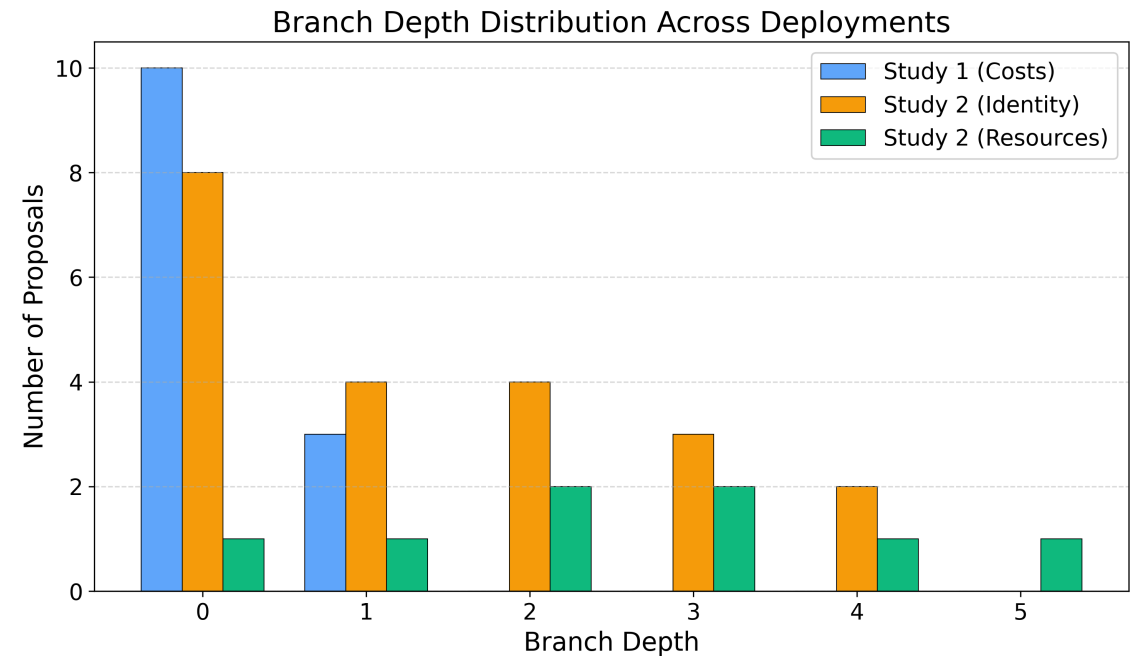
Both: real communities, real normative questions, German-language, 50–64% mobile access.

Mixed methods: platform telemetry · usability survey

Human Results: Emerging DAG Structures

Three distinct regimes emerged:

- **Flat Failure** (Study 1): Depth 1, 23% remix rate, 10 independent roots — virtually no building-upon
- **Deep Refinement** (S2-Q1): Depth 5, 87.5% remix, single iterative chain over 10 days
- **Balanced Synthesis** (S2-Q2): Depth 4, 62% remix, 6 multi-parent merges bridging thematic clusters



Human Results: Survey

On remixing difficulty:

- *"I wanted to comment on others' ideas rather than rewrite them, which I could not do"*
- *"It was difficult when multiple ideas were being discussed simultaneously"*
- *"With more participants it becomes increasingly difficult to reunite the common consensus when time is limited"*

On cognitive load & orientation:

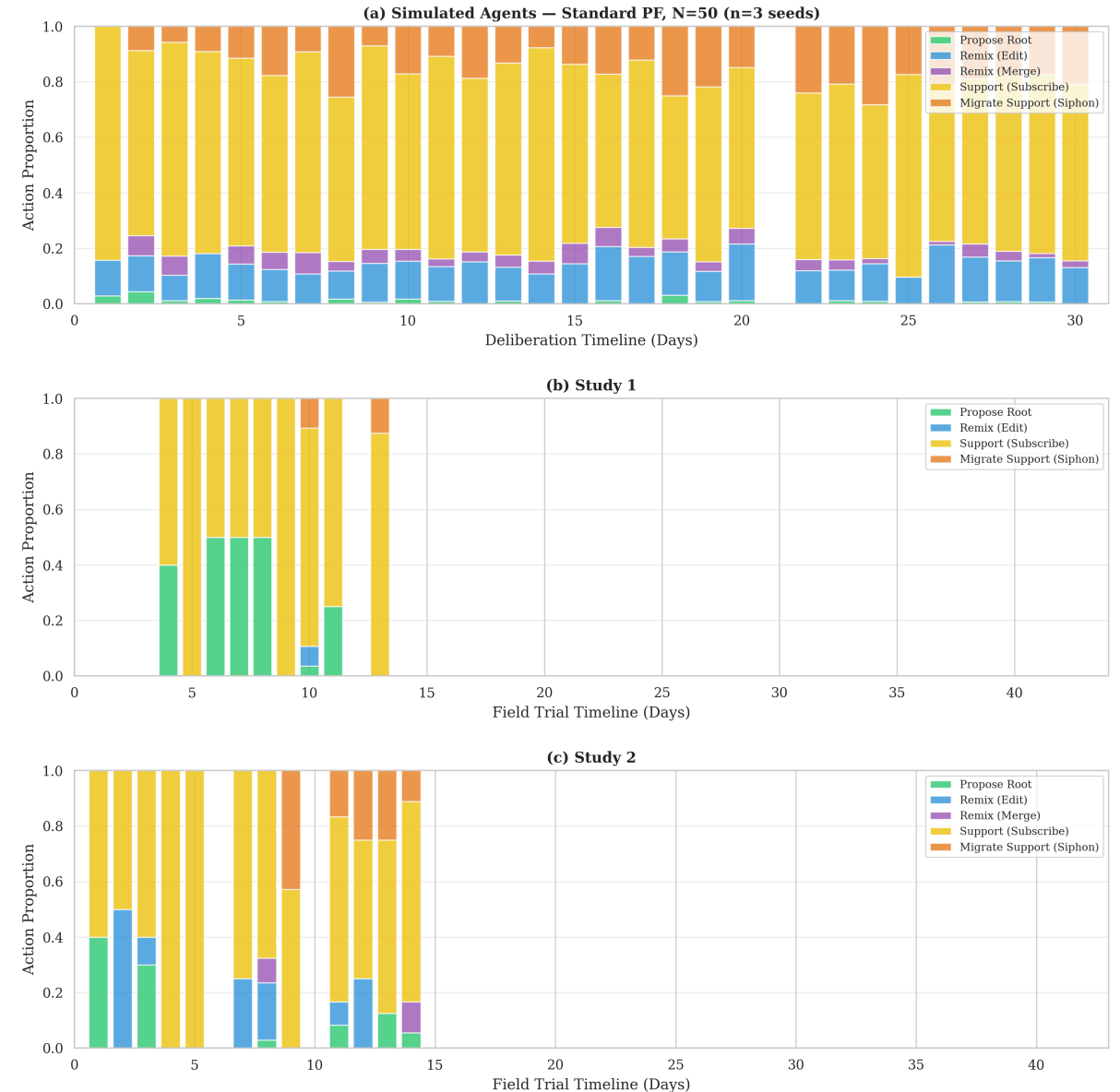
- *"The most difficult thing was to orient oneself"*
- SUS scores: **57.5** (S1) / **51.7** (S2) — both well below the 68 benchmark

On fairness & participation barriers:

- *"It favors motivated and engaged people with technical flair. You need to invest a certain start-up energy and return often to stay in the discussion"*

Human Results: Labels, Ballot, and Lessons

- **Labeling system:** Few users understood it. Labels propagate rigidly through the DAG — can't adapt when ideas evolve semantically.
- **Participation in later phases collapsed (Ballot voting):**
27% (S1) → 17% (S2)

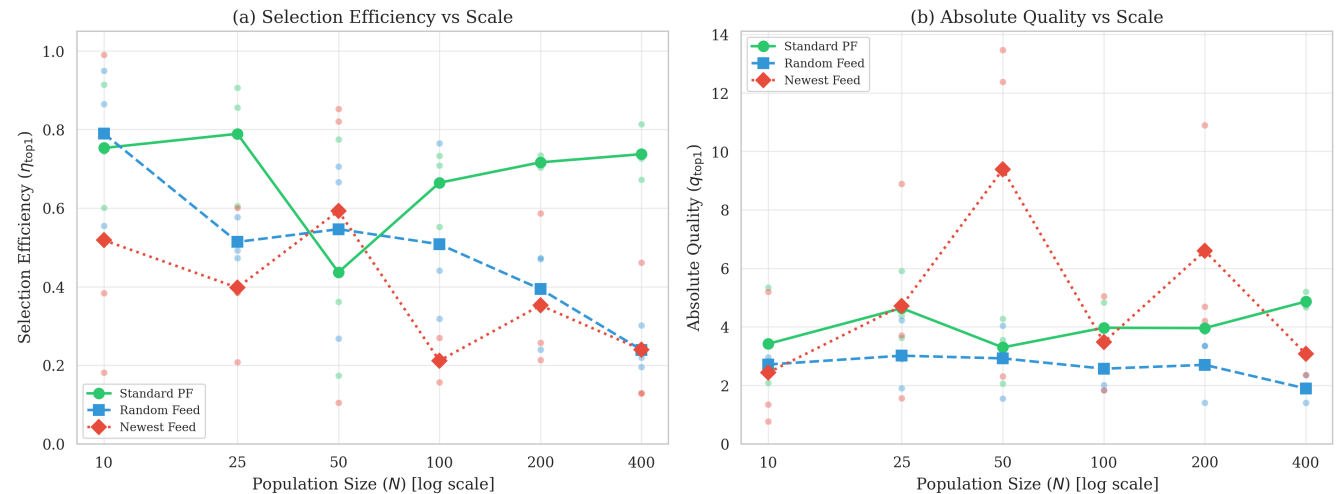


ABM Simulation: Setup

- Agents interact via the **actual platform API**
- **Bounded attention:** each agent sees at most **11 proposals** per session (3 Siphon + 3 Ballot + 5 PersonalFocus) — not global visibility
- **Quality Model:** User perceive a noisy quality score of a proposal. Remixing adds random zero mean perturbation on the quality
- **Asynchronous:** 30-day continuous timeline with Poisson sessions
- **Action Selection**
- **90 runs** across 3 suites: scaling ($N=10 \rightarrow 400$)

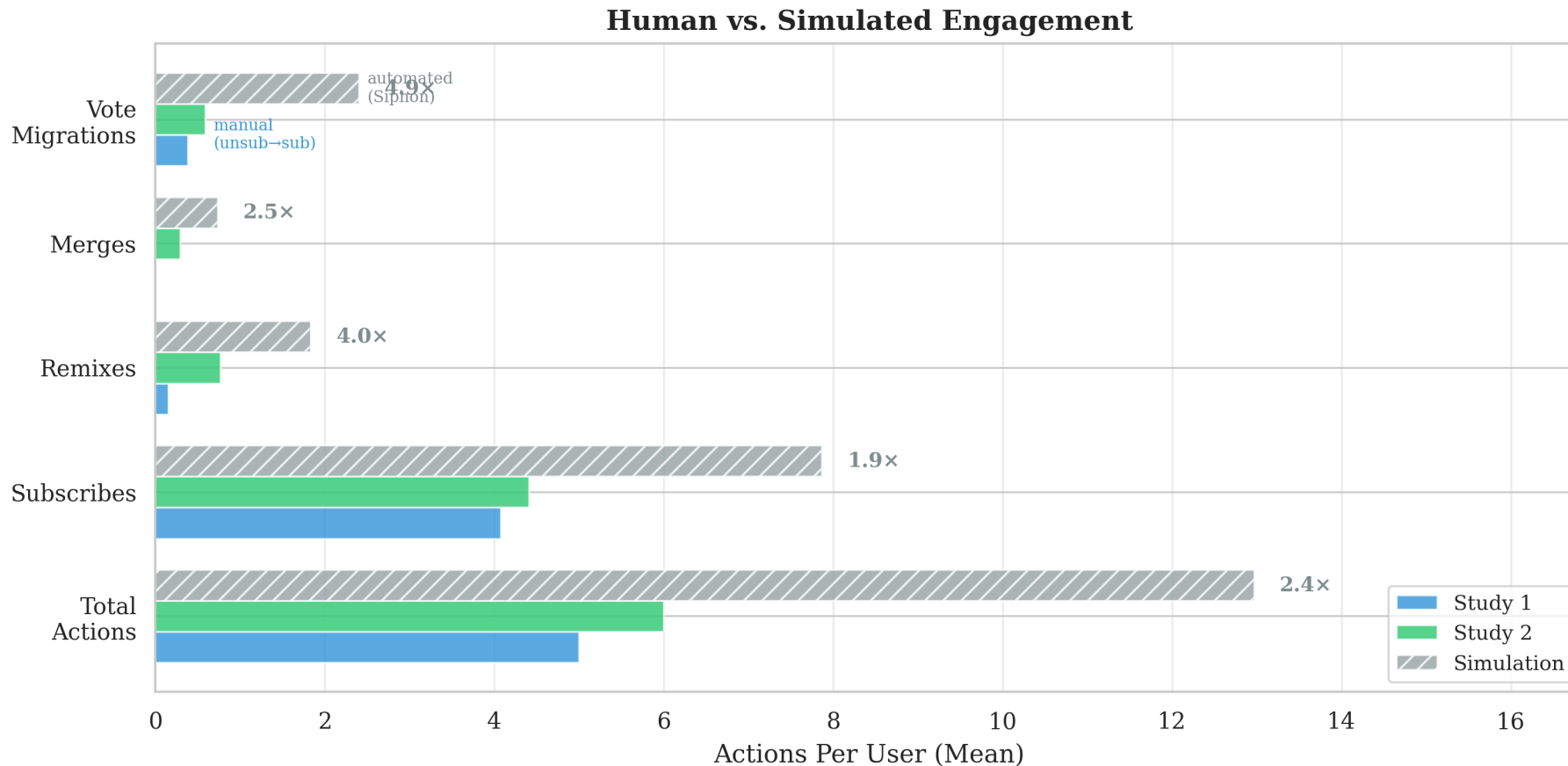
Simulation Results

- **Selection efficiency stays high:** PersonalFocus maintains $\eta > 0.70$ at $N=400$
- **Ballot decreases quality at scale:**
At $N=100$, unconstrained ranking $\rightarrow q = 4.88$
Label-exclusive ballot $\rightarrow q = 3.88$
- **Feed routing matters more** than ballot diversity for selection quality



Humans vs. Simulation

Agents do **2.4× more actions**, **4× more remixes**, and **1.9× more subscriptions** than humans.



The Core Asymmetry

Algorithmic Coordination

Scales ✓

Algorithms route attention and aggregate signals effectively
 $\eta > 0.70$ at $N=400$

Human Generation

Bottlenecked ✕

Checking changes and merging proposals is hard cognitive labour.

The selection machinery works. The generative machinery is bottlenecked by the cognitive friction of text communication communication.

Improvements for Next Iteration

- 📺 **Better (video) explanation of the system**
Most consistent user request across all studies. Terminology & Complexity was a major barrier.
- 💬 **Option to comment** — and potentially transform comments and questions into remixes
Bridging lightweight feedback with generative building-upon.
- 🖌️ **Reduce system complexity further:**
Label system might be ditched → simpler **DAG-lineage exclusivity** based on structural connection strength (via votes, subscriptions, even past subscriptions)
- 📱 **Native app with push notifications**
Better engagement than web-only. In-app notifications worked (56% response rate in Study 2), but a dedicated app would lower the entry barrier.

Outlook

Multimodal remixing:

Voice, drawings, video as input modalities → lower barriers for non-writers. Cross-modal translators could convert spoken or visual contributions into the text and vice versa.

The bigger vision:

If such a system can prove to work well at scale, it could replace other collective intelligence systems currently managing society.

Replacing chaotic and undemocratic ways of resource allocation by markets with **fine-grained democratic economic planning**: The historical bottleneck was informational (Cockshott & Cottrell), not computational. Synesis provides st exactly that.

Conclusion

"Although bottlenecked by the cognitive friction of communication, the DAG architecture provides the structural foundation to scale democratic collective intelligence."

"Lowering these transmission barriers — through better onboarding, lightweight interaction modes, or AI-assisted authoring — would unlock mass participation, extending democratic control into traditionally undemocratic domains."

Thank you.

Questions & Discussion

